University of BRISTOL

Centre for Academic Language and Development

# Foundations of Computer Programming - LANG00052

## Week 02

### Computational Thinking

Syed Murad

# Learning outcomes

- Developing a Comprehensive Understanding of Core Computational Thinking Concepts
- Understand Problem Decomposition
- Understand Pattern Recognition
- Understand Abstraction
- Understand Algorithm Thinking

# Computation Thinking

- Approaching a complex problem in a systematic manner
- Creating and describing a solution to a problem

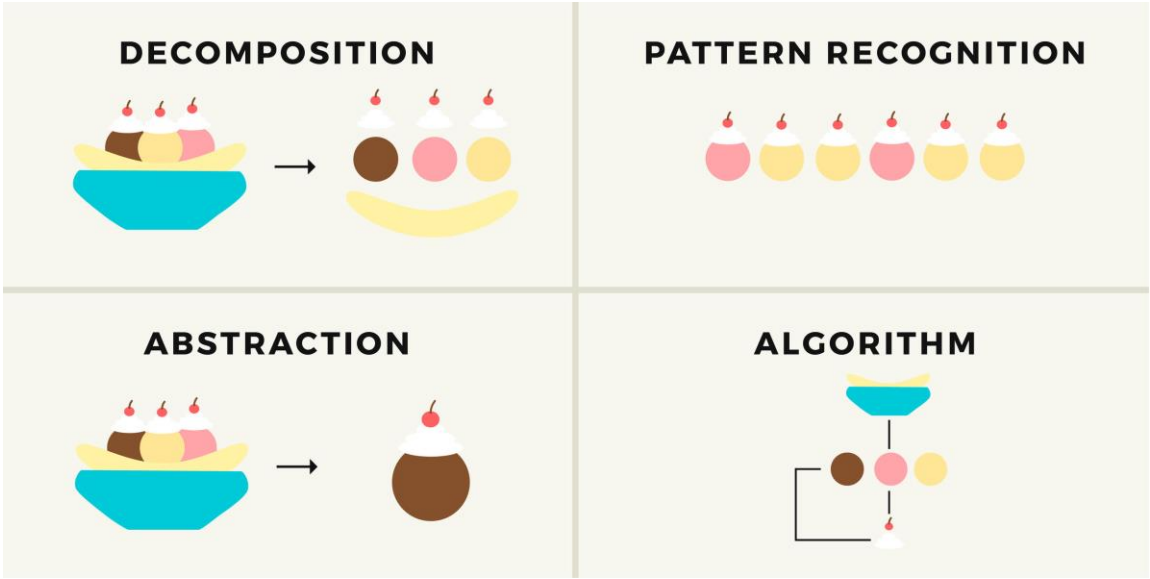
**Problem solving skill**

- Logical Thinking
- Critical Thinking

- ➢ Formulating a problem
- ➢ Expressing its solution

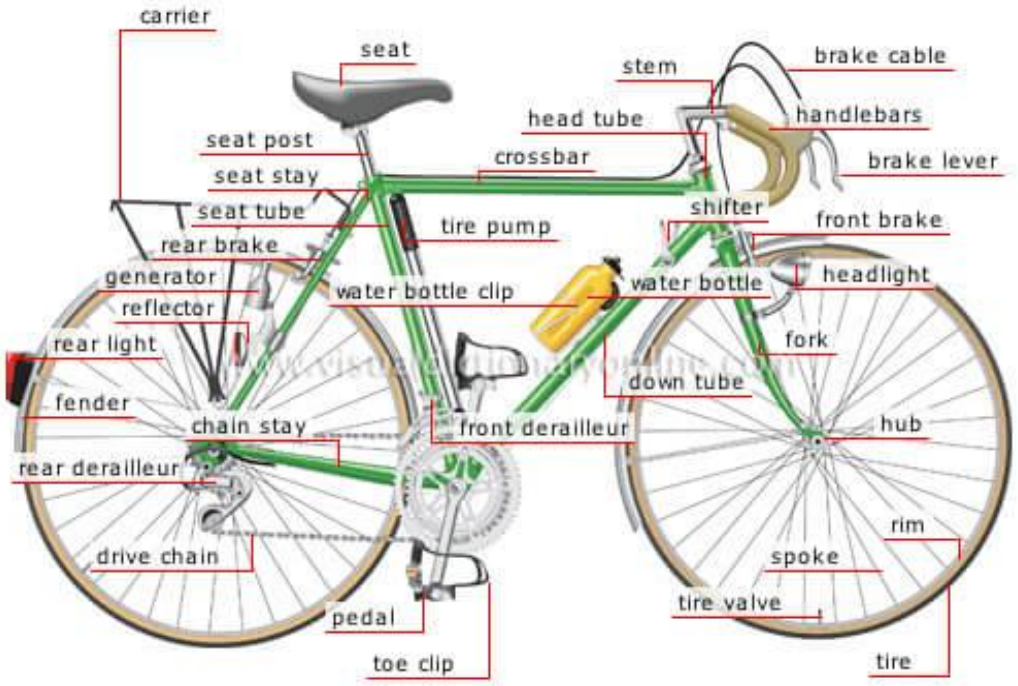
# COMPUTATIONAL THINKING TECHNIQUES

- **Decomposition**
- **Pattern recognition**
- **Abstraction**
- **Algorithmic thinking**



# Decomposition

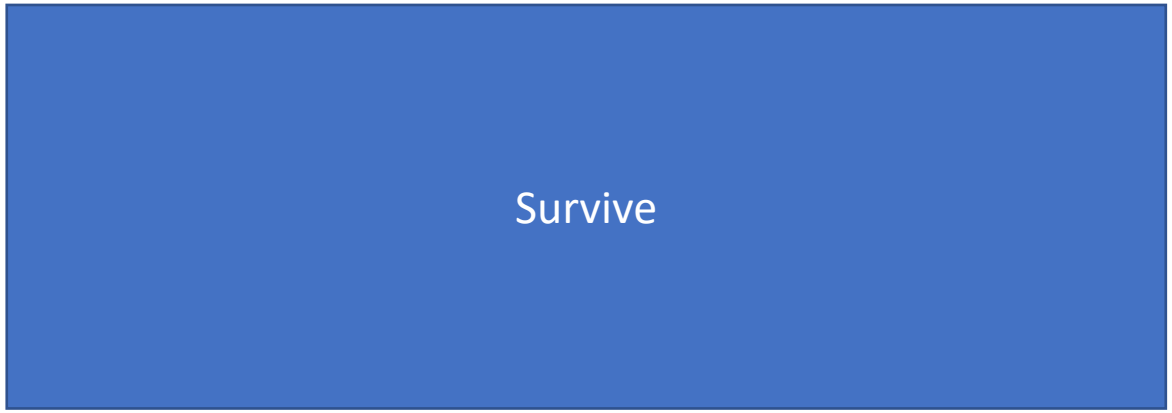
- Breaking down a complex problem into smaller parts that are more manageable and easier to understand.
- Those smaller parts can be examined and solved individually making the work simpler.

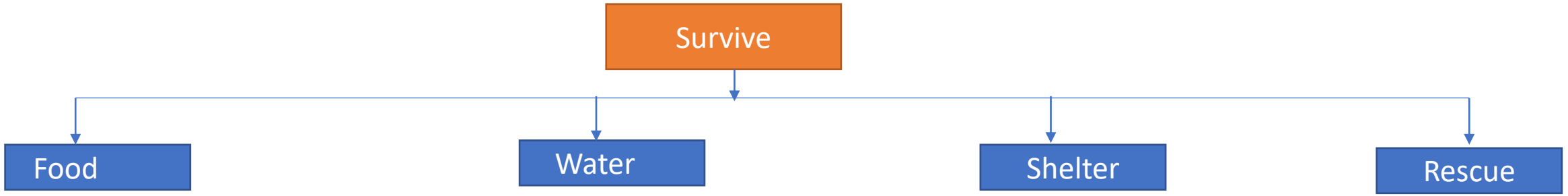




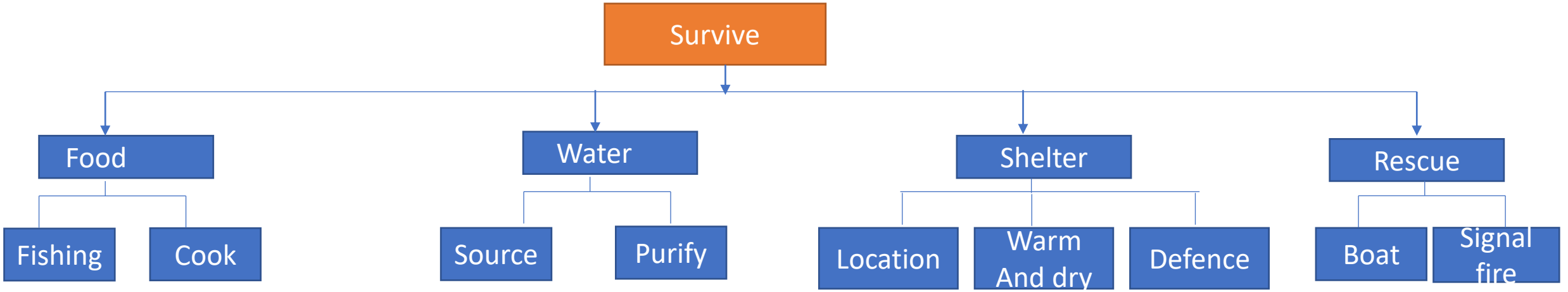
# I’m a student get me out of here

# Complex problem

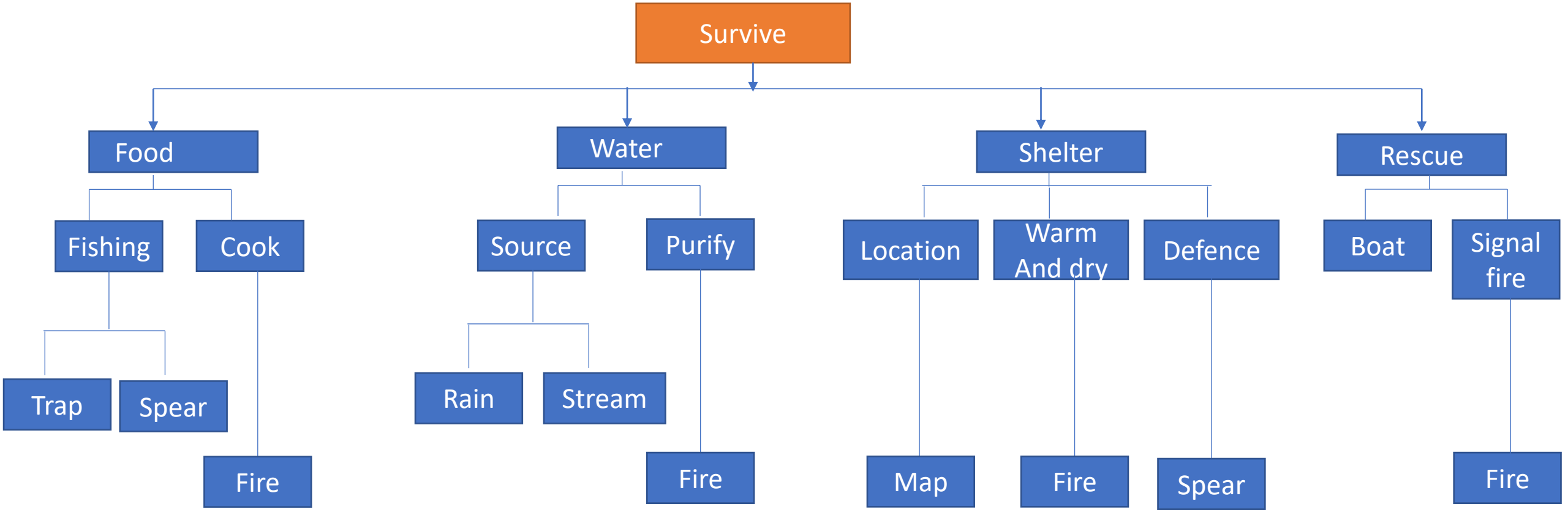



Survive
Food
Water
Shelter
Rescue

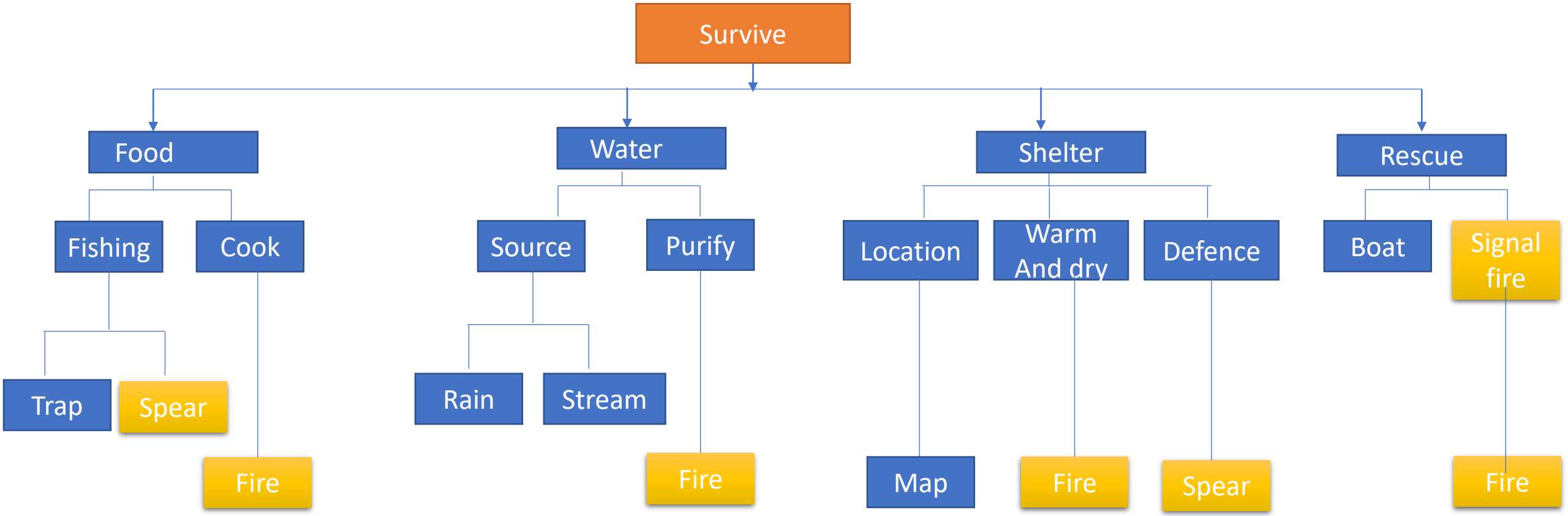
# DECOMPOSITION



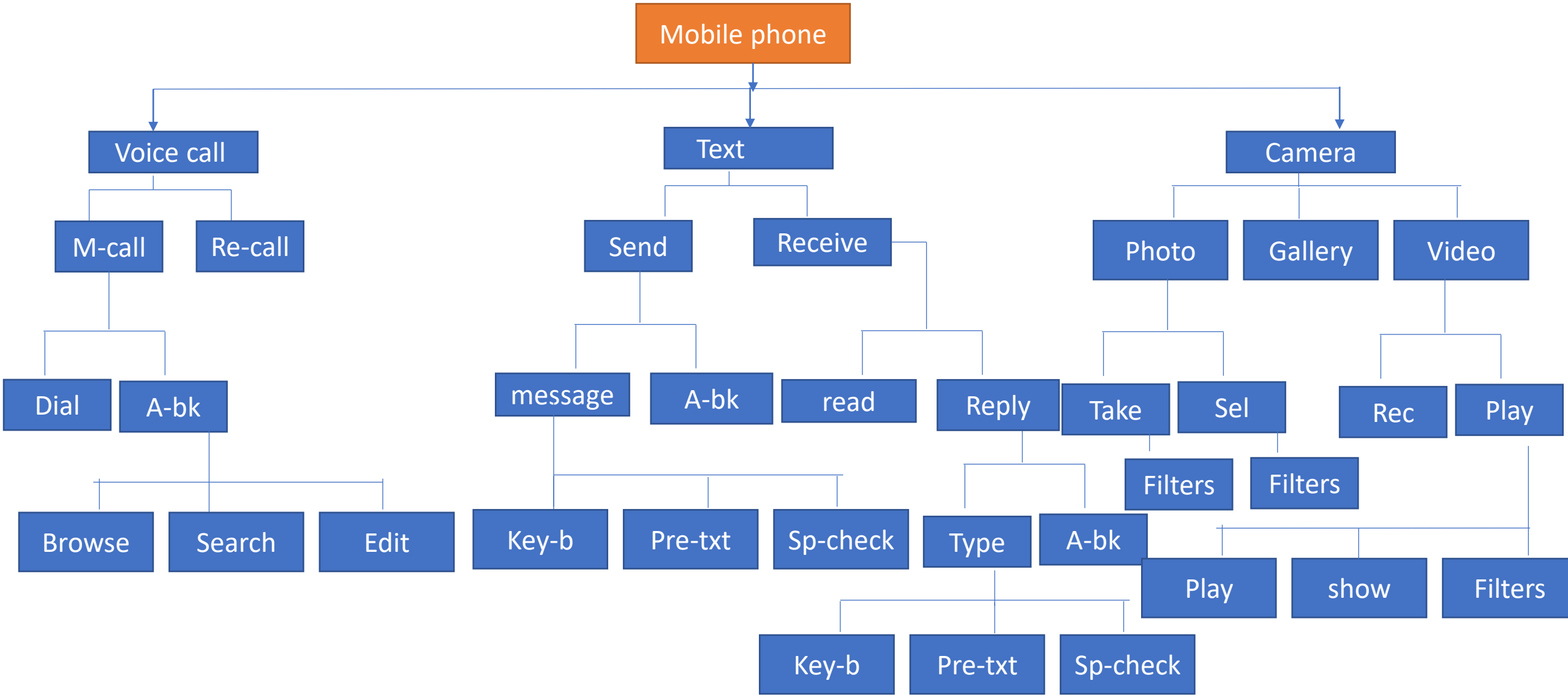
# DECOMPOSITION



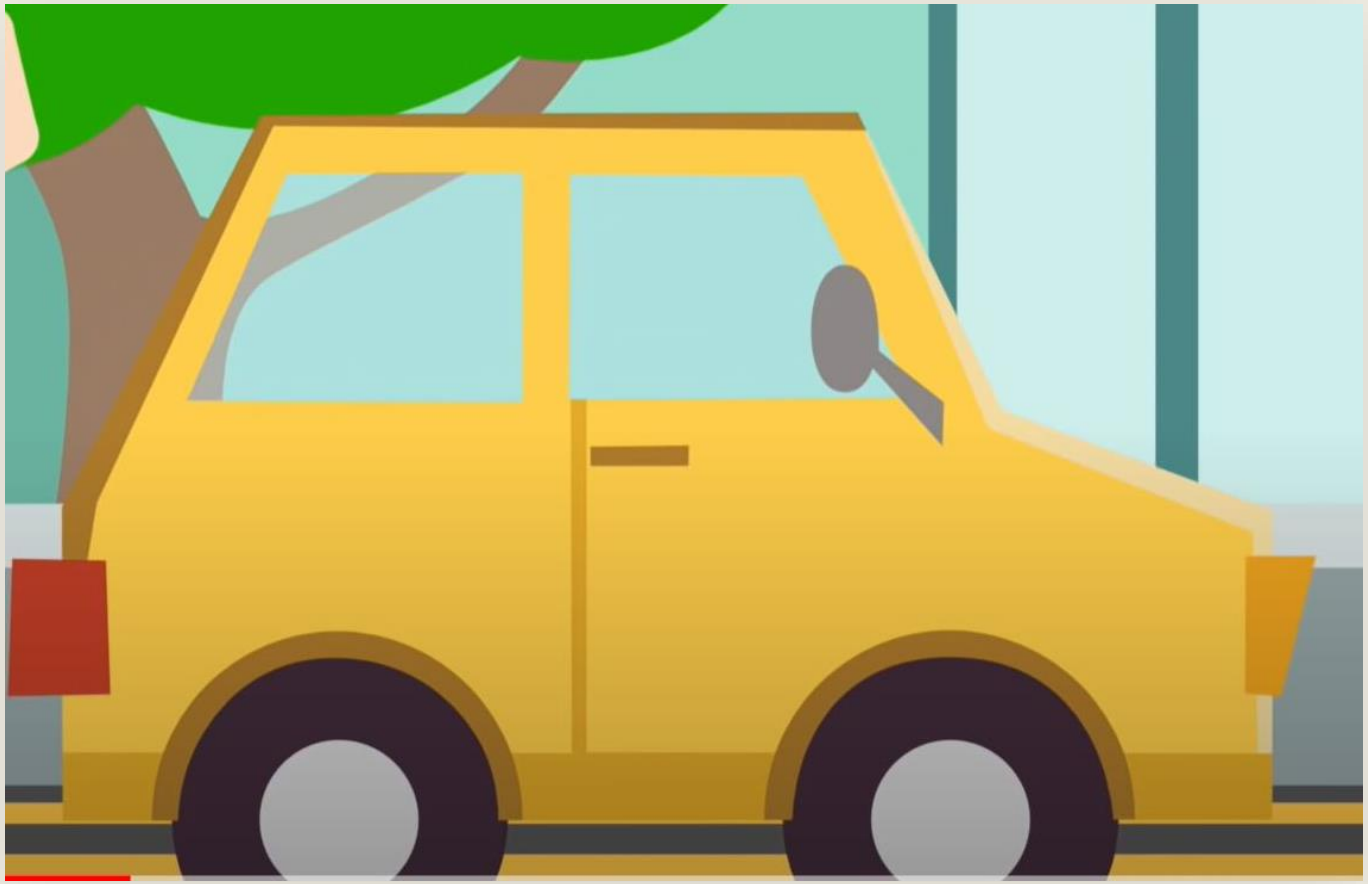
# DECOMPOSITION- Structural Diagram



# FUNCTIONAL DECOMPOSITION



# Exercise

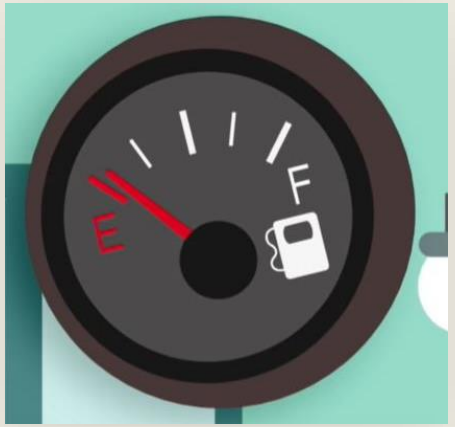




# DECOMPOSITION










# PATTERN RECOGNITION

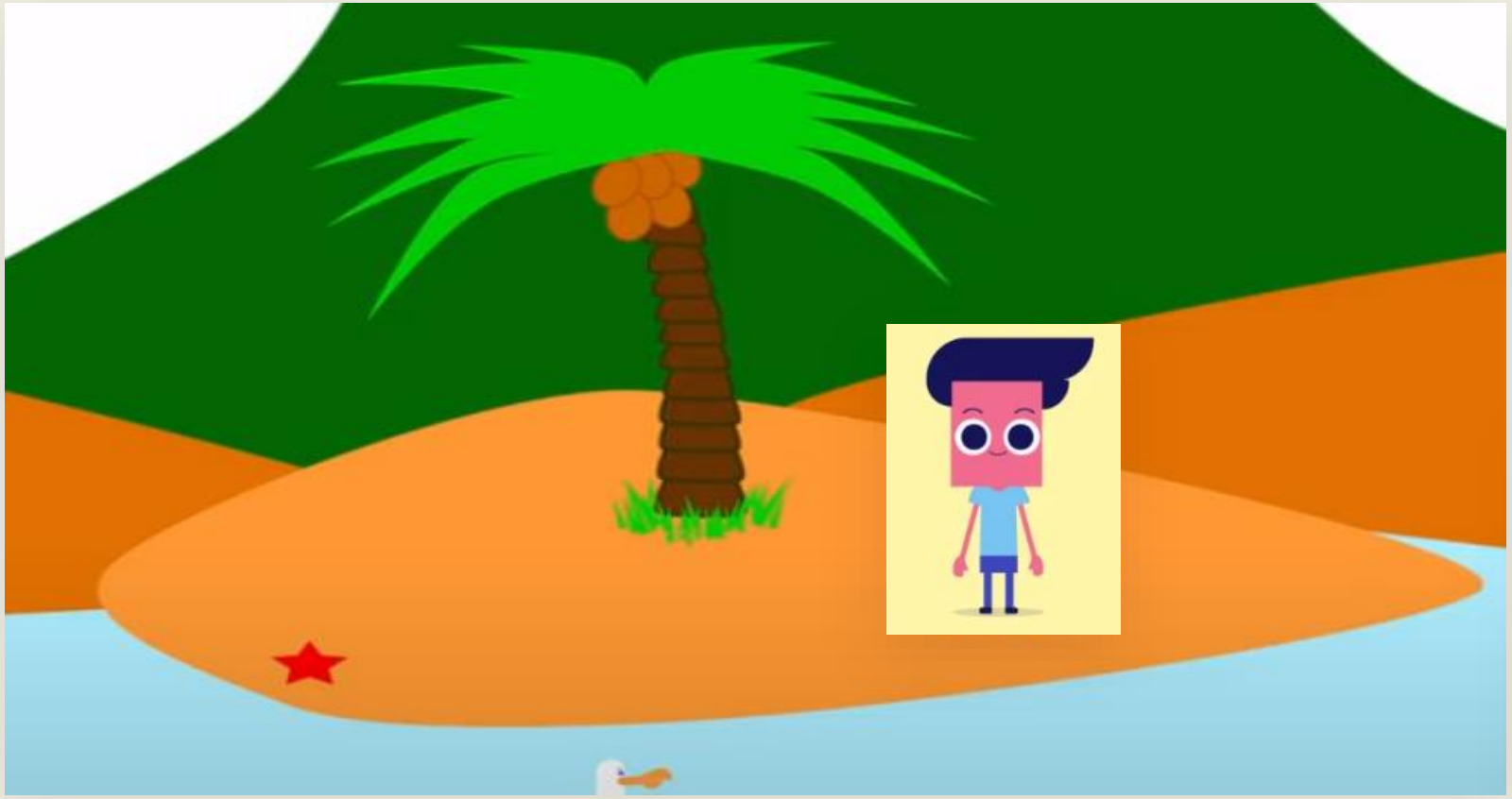
Pattern Recognition means recognizing the problem you need to solve is similar to another problem that is been solved before then apply the same or similar solution.

Example: A system developed to “Search for a Student”.

Aim: To search for students in a class, or who are being taught by a specific teacher.

Explanation: These both involve some form of “searching”, only difference is what is being searched.

# PATTERN RECOGNITION





# Exercise - PATTERN RECOGNITION





# Exercise - PATTERN RECOGNITION



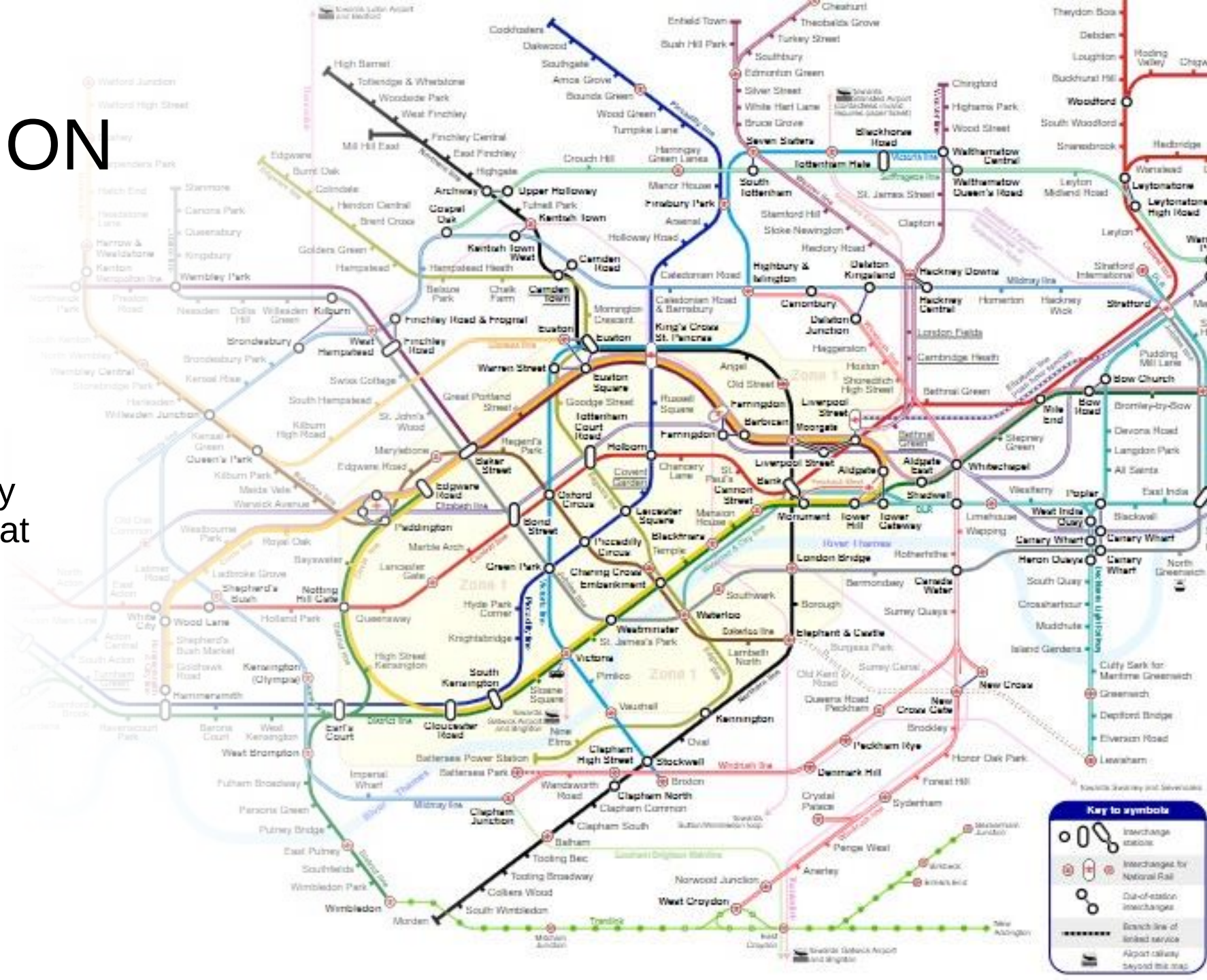






# ABSTRACTION

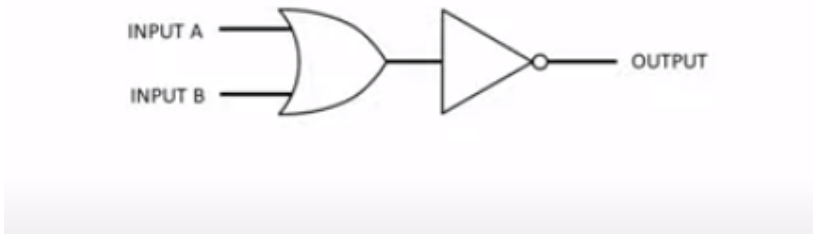
Abstraction is a powerful computational thinking technique that means simplifying a problem only focusing on the details that matter.



# ABSTRACTION

- X = 10
- Y = 20
- Z= X+Y

- Print(“Z”)





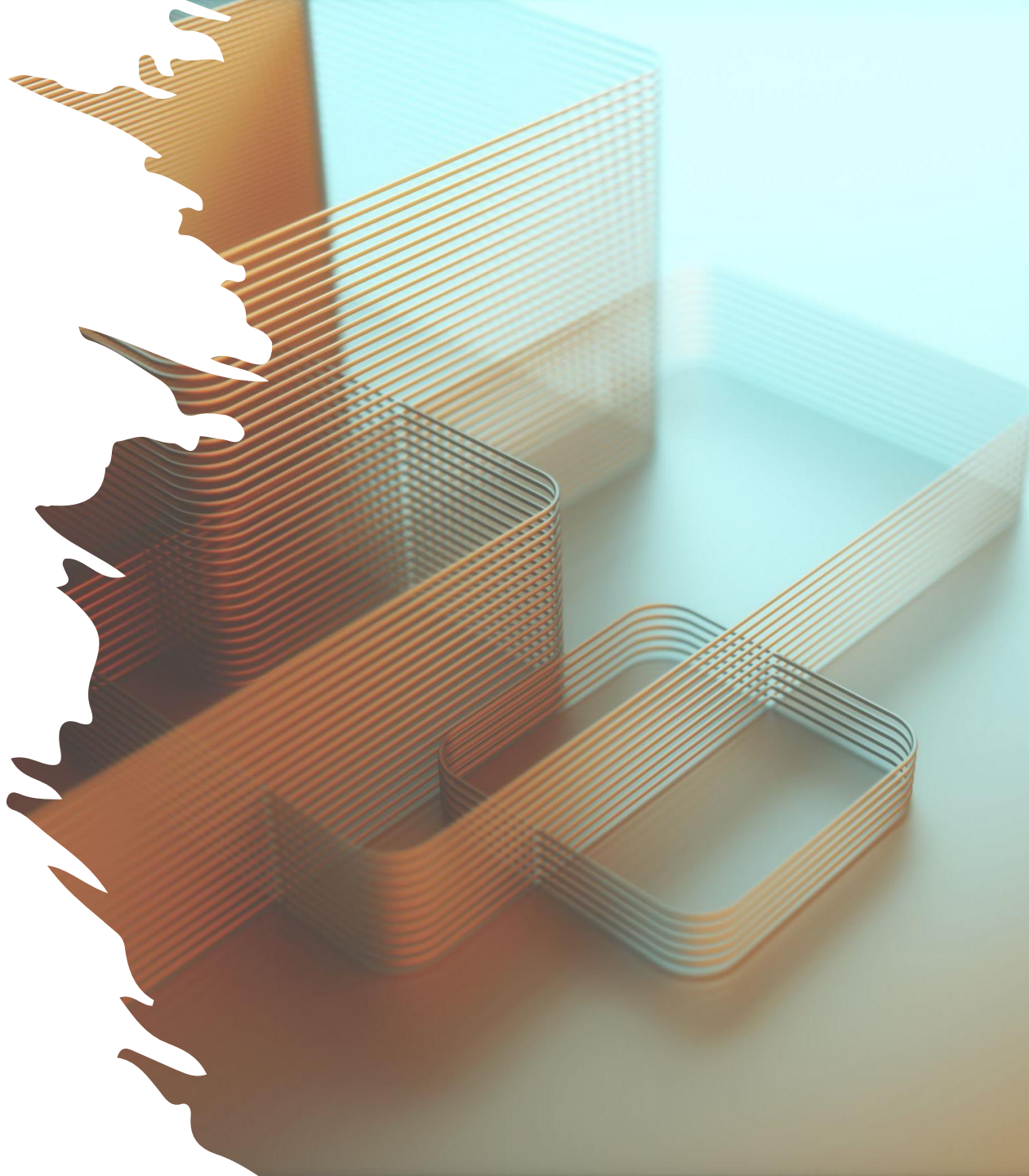

# ALGORITHMS

Set of rules or steps

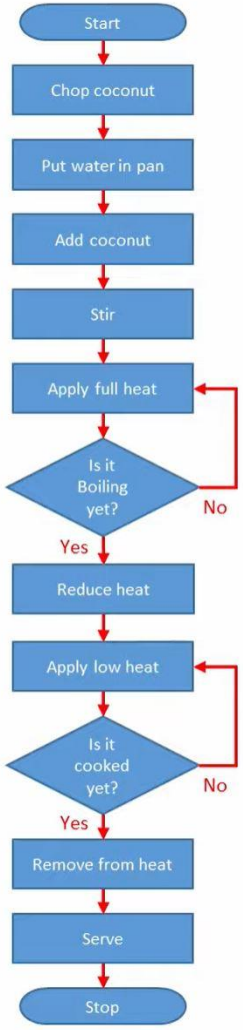
The most important techniques of computational thinking. An algorithms is another name for a set of steps or set of rules or both.

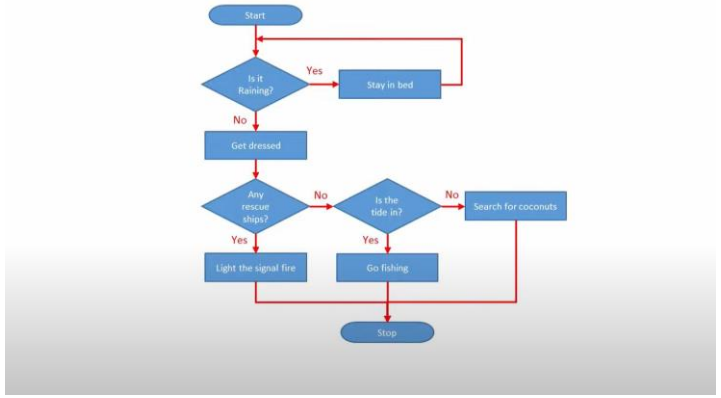
Computer scientists often use flowcharts to describe algorithms making it clear how one steps comes after another.

Algorithms often include logical reasoning.

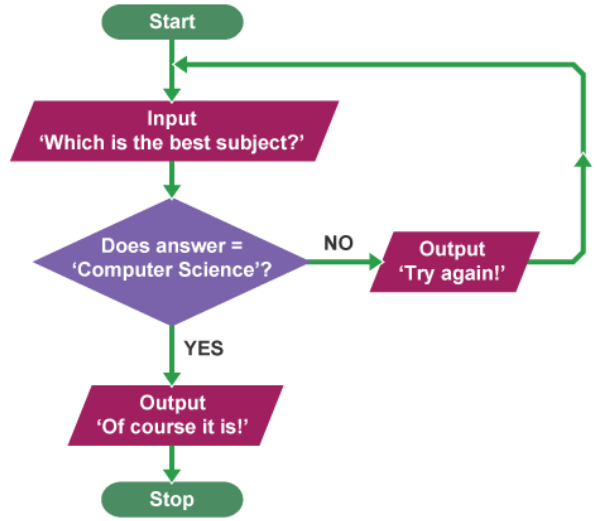


# ALGORITHMS


Start
Chop coconut
Put water in pan
Add coconut
Stir
Apply full heat
Is it Boiling yet?
No
Yes
Reduce heat
Apply low heat
Is it cooked yet?
No
Yes
Remove from heat
Serve
Stop

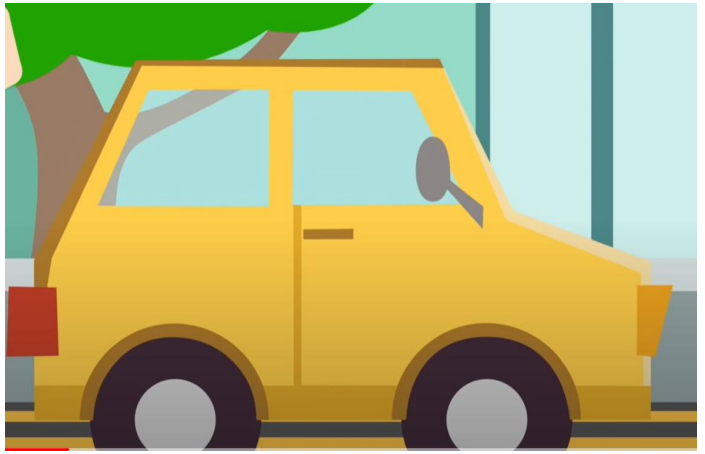

Start
Is it Raining?
Yes
Stay in bed
No
Get dressed
Any rescue ships?
No
Is the tide in?
No
Search for coconuts
Yes
Yes
Light the signal fire
Go fishing
Stop

# ALGORITHMS




```
REPEAT
    OUTPUT 'What is the best subject you take?'
    INPUT user inputs the best subject they take
    STORE the user's input in the answer variable
    IF answer = 'Computer Science' THEN
        OUTPUT 'Of course it is!'
    ELSE
        OUTPUT 'Try again!'
UNTIL answer = 'Computer Science'
```







# Exercise

# *Time for questions*